PART 1

1

EXPERIMENT 1

Identification of Halide Ions (Cl^-, Br^- and I^-)

Aim

To identify the presence of chloride, bromide and iodide ions from the given salt solutions.

Requirements

Test tubes, a test tube stand, a dropper, a beaker, etc.

Chemicals Required

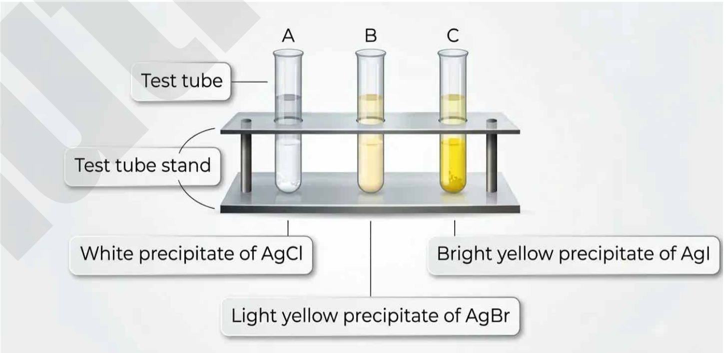
Silver nitrate solution ($AgNO_3$), and separate solutions of potassium chloride, potassium bromide and potassium iodide.

Procedure

1. Take three clean test tubes and mark them as A, B and C.
2. Pour about 5 ml of potassium chloride solution into test tube A, potassium bromide solution into test tube B, and potassium iodide solution into test tube C.
3. Add 4 to 5 drops of silver nitrate solution to each of the three test tubes and mix well.
4. Note down the colour of the precipitate that forms in each test tube.
5. Using the colour of the precipitate obtained, determine which halide ion is present in each solution.

Diagram

(Label the components in the diagram below)



Identification of halide ions (Cl^-, Br^- and I^- ions)

Observation Table

Test Tube	Reaction / Test	Colour of Precipitate	Inference

A	$KCl + AgNO_3$	White	Chloride (Cl^-) present.
B	$KBr + AgNO_3$	Light yellow	Bromide (Br^-) present.
C	$KI + AgNO_3$	Bright yellow	Iodide (I^-) present.

Chemical Reactions

(1) $KCl\ (aq) + AgNO_3\ (aq) \rightarrow KNO_3\ (aq) + AgCl\ (s)\downarrow$
(Potassium chloride) + (Silver nitrate) → (Potassium nitrate) + (White precipitate of silver chloride)

(2) $KBr\ (aq) + AgNO_3\ (aq) \rightarrow KNO_3\ (aq) + AgBr\downarrow$
(Potassium bromide) + (Silver nitrate) → (Potassium nitrate) + (Light yellow precipitate of silver bromide)

(3) $KI\ (aq) + AgNO_3\ (aq) \rightarrow KNO_3\ (aq) + AgI\downarrow$
(Potassium iodide) + (Silver nitrate) → (Potassium nitrate) + (Bright yellow precipitate of silver iodide)

Inferences

1. Halide ions (Cl^-, Br^-, I^-) are precipitated in all the three reactions of above experiments.
2. In the above experiment, depending on the colour of the precipitates, presence of Cl^-, Br^- and I^- in the given solutions are confirmed.
3. Elements of halogen group in the periodic table show similar properties.

Multiple Choice Questions

1. The number of electrons in the outermost shell of halogens is ....
(A) 1 (B) 2 (C) 3 (D) 7
Answer: (D) 7

2. .... is in solid state at room temperature.
(A) Fluorine (B) Iodine (C) Chlorine (D) Bromine
Answer: (B) Iodine

3. Which of the following substances is the strongest reducing agent?
(A) Cl_2 (B) Cl^- ion (C) Br (D) Br^- ion

Answer: (D) Br^- ion

Remember

1. All halogen elements are oxidizing agents.
2. Precipitation reactions produce insoluble salts.

Date: ______________ Teacher's Signature: ______________